

Stat 992-001: Statistical methods for QTL mapping (Spring, 2011)

Homework #4: Mapping multiple QTL

Due 13 Apr

Grab the comma-delimited data file at

<http://www.biostat.wisc.edu/~kbroman/teaching/qtltopics/hw4.csv>

This is simulated data from a backcross with 300 individuals, with a single quantitative phenotype.

Perform 1- and 2-dimensional QTL scans, exploratory multiple QTL analysis as well as the automated multiple-QTL mapping procedure in `stepwiseQTL` to identify QTL and QTL×QTL interactions with these data.

Write a short report describing what you did and what you found, including the relative support for different QTL and QTL×QTL interactions.

You can use Haley-Knott regression for all analyses, to save computation time.

Also, I've run permutation tests with 1- and 2-dimensional scans and placed the results at the following:

<http://www.biostat.wisc.edu/~kbroman/teaching/qtltopics/hw4.RData>